

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			- photosynthesis produces {oxygen/ O₂} / {oxygen/ O₂} is a bi-product of photosynthesis (1) Not O² or o² Reject equation on own - Therefore if the production of O₂ {increases/decreases}/ if there is {more/less} O₂ (1) (The rate of) photosynthesis is {increasing/decreasing}. (1) The more oxygen the more photosynthesis = 2 marks This question relates to the 'rate of photosynthesis' and not to whether the plant is photosynthesising or not		3		3		3
	(b)			Any 1 from: - Increasing light intensity has no effect on O₂ production/photosynthetic rate - Increasing CO₂ concentration has no effect O₂ production/ photosynthetic rate (1) - Increasing temperature increases O₂ production/ photosynthetic rate (1) Any 1 from: - Therefore temperature must be the <u>limiting factor</u> - the temperature is <u>too low</u> to increase O₂ production/ photosynthetic rate (1)			2	2		2

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	(c)			CO ₂ was the limiting factor (1) Any 1 (x1) from: • {Increasing/ change in} temperature has no effect on O₂ production/ photosynthetic rate • {Increasing/ change in} light intensity has no effect on O₂ production/photosynthetic rate • Only when carbon dioxide concentration increases does the photosynthetic rate increase			2	2		2
	(d)			To prevent {gases/ air/ oxygen/ carbon dioxide} {entering/leaving}		1		1		1
	(e)			<i>Factor</i> – light in the room/ light around the apparatus/ natural light OR temperature outside the container/room temperature (1) <i>How factor could be controlled –</i> LIGHT – place in dark/ black out container/make the container light proof/carry out expt in a (light proof) cupboard (1) NOT turning lights off in room/ opening windows to adjust light OR TEMPERATURE – container needs thermostatic control/ OWTTE		1	1	2		2
				Question 6 total	0	5	5	10	0	10